Employee Payroll and Deduction Analysis

Complete worked solution

Sample payroll policy: gross salary = basic + allowances + overtime pay + incentive. Deduct 0% on the first 30000, 10% on the next 30000, and 20% above 60000 (marginal bands). The sample overtime rate is 1.5 times basic hourly rate, with 160 regular hours/month. Print a formatted statement for each employee.

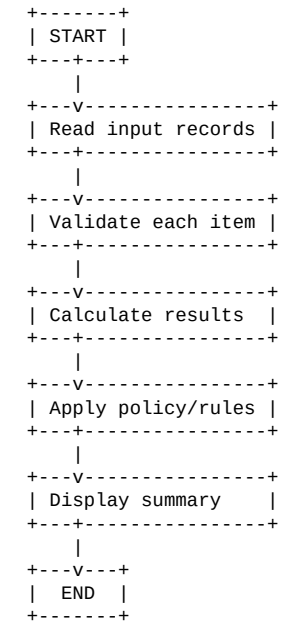
1. Computational Thinking and Algorithm

Inputs: Supplied records and policy values. **Processing:** Validate data, calculate charges or scores, apply ordered rules, and compare outcomes. **Outputs:** Itemized results, classifications, rankings, or recommendations.

Pseudocode

```
START
  FOR each employee
    Validate salary components and hours
    Calculate hourly rate from basic monthly pay / 160
    Calculate overtime pay and gross salary
    Apply progressive marginal deductions
    Net pay = gross - deduction
    Display formatted statement
END
```

2. Flowchart



3. Python Program

The program uses basic variables, numeric and string data, tuples, lists, loops, conditionals, arithmetic, and formatted output. Policy values are constants near the top of the program.

```
employees = [
    ("E301", "Ravi", 52000, 8000, 12, 5, 4000),
    ("E302", "Nila", 38000, 6000, 8, 0, 1500),
]
# id, name, basic, allowances, overtime_hours, incentives, other_bonus
standard_hours = 160
for emp_id, name, basic, allowances, overtime_hours, incentives, bonus in employees:
    components = [basic, allowances, overtime_hours, incentives, bonus]
    valid = True
```

```
for amount in components:
    if amount < 0:
        valid = False
if not valid or standard_hours <= 0:
    print("Invalid payroll record:", emp_id)
    continue
hourly_rate = basic / standard_hours
overtime_pay = overtime_hours * hourly_rate * 1.5
gross = basic + allowances + overtime_pay + incentives + bonus
band1 = min(gross, 30000) * 0.00
band2 = max(0, min(gross - 30000, 30000)) * 0.10
band3 = max(0, gross - 60000) * 0.20
deduction = band1 + band2 + band3
net = gross - deduction
print("=" * 48)
print("SALARY STATEMENT -", name, "(", emp_id, ")")
print("Basic pay:       ", format(basic, ".2f"))
print("Allowances:      ", format(allowances, ".2f"))
print("Overtime pay:    ", format(overtime_pay, ".2f"))
print("Incentives/bonus:", format(incentives + bonus, ".2f"))
print("Gross salary:    ", format(gross, ".2f"))
print("Deduction:       ", format(deduction, ".2f"))
print("NET PAY:         ", format(net, ".2f"))
```

4. Sample run guidance

Run the program as written to see a sample result. Replace the sample lists and constants with the values provided in your assignment. Keep the rule order and units consistent; for example, percentages should be entered as percentages and converted to fractions only in the calculation.